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META STRUCTURE SEARCH FOR LINK WEIGHT PREDICTION IN HETEROGENEOUS GRAPHS

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ABSTRACT

Recently link weight prediction has attracted an increasing research interest due to its merits in quantifying the strength between nodes within a graph. Nonetheless, current link weight prediction methods focus solely on graph topology, disregarding node feature information embedded in graphs. In real-world applications, we often collect heterogeneous graph data where multiple types of nodes linked by multiple types of edges are available for analysis, and it is essential and challenging to quantify the proximity of different types of nodes. To solve this challenge, we present a new model for **H**eterogeneous Graph **L**ink **W**eight **P**rediction (HLWP for short). In HLWP, message passing in heterogeneous graph neural networks is described as a meta structure, which can be effectively designed by Differentiable Neural Architecture Search (DARTS) algorithms. Thus, HLWP can enhance the message passing in heterogeneous graphs by DARTS. In addition, HLWP employs a perturbation-based algorithm to enhance stability and precision. Through empirical experiments conducted on three real-world datasets, we demonstrate that HLWP achieves accurate predictions of link weights. Our results highlight the superiority of HLWP over existing methods for link weight prediction and baseline GNN models in terms of accurately predicting link weights within heterogeneous graphs.

Index Terms— link weight prediction, heterogeneous graph, meta structure, graph neural networks

1. INTRODUCTION

Link prediction plays a fundamental role in graph analysis, which aims to predict the presence of connections between nodes in a graph [1, 2]. Link prediction has wide-ranging applications in various fields [3, 4], such as studying social networks, completing knowledge graphs , analyzing traffic networks, and reconstructing biochemical domains. As a branch of link prediction, link weight prediction has attracted an increasing interest, where links are assigned weights representing their strength or importance and the weights can describe distance or similarity between nodes, providing valuable information for popular tasks like optimizing traffic and social interactions understanding. For instance, link weights can indicate the level of interaction, similarity in social networks, and the volume of traffic in transportation systems.

Existing research works use traditional machine learning models to tackle link weight prediction. For instance, Fu et al. [5] uses statistical features extracted from graphs through edge and node swapping, employing machine learning techniques like random forest for edge weight forecasting. With the rise of deep learning, a variety of deep learning methods have been developed for link prediction. ModelR [6] uses known link weights to derive node attributes, passing them through two hidden layers, and predicting unknown link weights. However, ModelR's effectiveness was hindered by the randomness in node sorting during feature extraction. LWP-WL [7] solves the problem of not paying attention to the order of nodes in ModelR and directly putting it into the neural network (CNN). LWP-WL believes that the nodes put into the CNN should not be simply drawn directly from the neighbors of the nodes, but should have a certain order. For example, the nodes that are closer to the topology of the nodes at both ends of the target should be placed in the front. Therefore, the WL-test algorithm [8] is used to reorder the topological node set of nodes, and then put them into CNN for training.

The above models achieve substantial progress in link weight prediction. However, they overlook node types in heterogeneous graphs and fail to capture intricate semantic data in heterogeneous graphs [9, 10]. Recently, a range of models based on Graph Neural Networks (GNNs) have emerged to leverage the nuanced meanings within heterogeneous graphs [11, 12, 13, 14]. These models have made significant strides in various graph-based machine learning tasks, such as node classification and link prediction. For instance, HAN [15] and MAGNN [16] incorporate domain-specific insights to design appropriate meta paths, guiding the creation of node representations for downstream tasks. GTN [17] use self-attention to capture dynamic relationships among nodes in heterogeneous graphs. Extracting more complex relationships from topological structures using meta paths requires substantial effort, as meta paths inherently capture linear information. As a result, certain models integrate meta structures [18] into heterogeneous graph representations to gather additional topological insights. For example, DiffMG [19] devises a set of candidate meta structures and employs differentiable neural architecture search (DARTS [20]) to identify the most suitable structure. We apply these methods to link weight prediction in heterogeneous graphs and the experimental results in Table 2 show that the meta structure based method for link weight prediction obtains better performance compared with previous ones. The main reason for this advantage is that the link weight prediction task pays more attention to the local topological structure and semantic features of a link when predicting its weight. For example, when we predict the traffic flow of a road, we should not only care about the traffic information of the road's two ends, but should pay more attention to the overall traffic flow information of the local traffic network [21]. Obviously, meta structure based methods are more advantageous when dealing with this type of task.

In this paper, we propose a new perturbation based meta struc-

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ture selection approach, termed as HLWP (**H**eterogeneous Graph **L**ink **W**eight **P**rediction), to solve the link prediction task. Our motivation stems from the observations that when predicting the link weight under the framework of meta structure based GNN method, each message passing step should be beneficial for the final prediction. So when we select the message passing mode, it should reflect the gain of this step for link weight prediction. However, the existing works select the message passing mode via the mode's parameters and ignore the impact on the final task, limiting further improvement in prediction performance. Under this observation, we propose an alternative perturbation-based meta structure selection that is able to measure the influence of message passing modes directly and select the most influenced one as the message passing mode. The experimental results show that the proposed method achieves the best results in link weight prediction task so far.

Overall, HLWP involves four main steps. First, formulate message passing using meta structures for heterogeneous graphs. Then, train meta structure pair parameters through differentiable neural architecture search. Next, select the final meta structure pair using perturbation maximization for link weight prediction. At the last step, train a Graph Convolution Network (GCN) model and predict link weights using the chosen meta structure pair. Empirical evaluations on three real-world datasets demonstrate HLWP's superiority over previous models. Ablation studies confirm that HLWP performs best with its two key components. The main contributions can be summarized as follows:

1. This work represents the first effort to study the problem of link weight prediction in heterogeneous graphs, where a message passing model is designed based on meta structures. Our approach can represent the links for graphs in a higher dimension and thus achieves better results.
2. We develop a new model HLWP, which uses DARTS-based search and perturbation-based selection to generate the meta structures for links.
3. We evaluate the performance of HLWP on real-world graph datasets. The results show that HLWP performs better than existing GNN models and link weight prediction models.

2. PRELIMINARIES

2.1. Definitions

Definition 1 ***Heterogeneous Graph***. *A heterogeneous graph is defined as a graph $G = (V, E, T, R)$, where V is the set of nodes and E is the set of edges. Each node v and edge e are associated with their type mappings $v \rightarrow T$ and $e \rightarrow R$. T denotes the set of nodes' types, and R denotes the set of edges' types, where $|T| + |R| > 2$.*

Definition 2 ***Meta Structure*** *$m\mathcal{S}$. A meta structure $m\mathcal{S}$ is a graph with node in T and the edge relation in R. A meta path is a special case of a meta structure.*

Meta paths primarily emphasize the depiction of linear and simplistic relational paths among nodes within a graph, whereas meta structures can characterize more intricate and diverse structural patterns in the graph.

2.2. Problem Formulation

In this work, we aim to predict the link weight in the heterogeneous graph. Since the edge to be predicted lacks its attributes, we derive its features from nodes at both ends. Here, the two nodes are source node n_s (short for s) and target node n_t (short for t). Given a heterogeneous graph G and link weight task-related source node s and target node t, we aim to find the best meta structure pair ($m\mathcal{S}_s^*$, $m\mathcal{S}_t^*$) that minimize the loss ℓ. The problem can be formulated as below:

$$(m\mathcal{S}_s^*, m\mathcal{S}_t^*) = \min_{m\mathcal{S}_s \in \mathcal{S}_s, m\mathcal{S}_t \in \mathcal{S}_t} \ell(W_{pre}(m\mathcal{S}_s, m\mathcal{S}_t), W_{label}), \quad (1)$$

where W_{pre} is the predicted link weight with the meta structure pair ($m\mathcal{S}_s$, $m\mathcal{S}_t$). W_{label} is the actual link weight between the source and target nodes. $\mathcal{S}_s$ and $\mathcal{S}_t$ represent the set of all possible outcomes for the meta structure pair ($m\mathcal{S}_s$, $m\mathcal{S}_t$).

3. METHODOLOGY

The proposed HLWP is illustrated in Figure 1. In this part, we will introduce the model in detail.

3.1. Meta Structure Based Message Passing

A GNN learns node representations by aggregating information from nodes' neighbors via the message-passing scheme. This process can be expressed as follows:

$$\boldsymbol{H}^{(l+1)} = \mathrm{F}\left(\boldsymbol{H}^{(l)}, A\right), \quad (2)$$

where $\mathrm{F}\left(\boldsymbol{H}^{(l)}, A\right)$ denotes aggregate function $\boldsymbol{H}^{(l)}$ from neighbors defined by A. When $l = 0$, $\boldsymbol{H}^{(0)} = \mathrm{MLP}(X)$, X denotes the node features. Specifically, we choose GCN [22] as the basic GNN, which is the most popular GNN instance. $\boldsymbol{H}^{(l)}$ is the node representations in the l-th layer. However, for heterogeneous graphs, Equation (2) indiscriminately aggregates messages coming along all relations without discrimination, resulting in the loss of semantic information.

Therefore, we use an expressive meta structure $m\mathcal{S}$ to characterize the message-passing process between GNN layers by selecting edge types when aggregating information from neighbors. The first layer $\boldsymbol{H}^{(0)}$ represents the input and the last layer represents $\boldsymbol{H}^{(L)}$ the representations of target nodes. And other intermediate layers represent the intermediate representations in the computation process. L is the predefined number of total layers. In the meta structure, each layer $\boldsymbol{H}^{(l)} (1 \leq l \leq L)$ is computed based on all of its predecessors:

$$\boldsymbol{H}^{(l)} = \sum_{0 \leq i < l} f_{i,l}\left(\boldsymbol{H}^{(i)}, \mathcal{A}_{i,l}\right), \boldsymbol{H}^{(0)} = \mathrm{MLP}(X), \quad (3)$$

$f_{i,l}$ denotes passing $\boldsymbol{H}^{(i)}$ message along a particular relation type given by the meta structure, and this is selected from $\overline{\mathcal{A}}_{i,l}$. $\overline{\mathcal{A}}_{i,l}$ is defined as:

$$\overline{\mathcal{A}}_{i,l} = \begin{cases} \mathcal{A} \cup \{I\} & i = l-1 \text{ and } l < L \\ \mathcal{A} \cup \{I\} \cup \{O\} & i < l-1 \text{ and } l < L \\ \overline{\mathcal{A}} & i = l-1 \text{ and } l = L \\ \overline{\mathcal{A}} \cup \{I\} \cup \{O\} & i < l-1 \text{ and } l = L \end{cases}, \quad (4)$$

where $\mathcal{A}$ is the adjacency matrix set of the heterogeneous graph G, $\overline{\mathcal{A}}$ is the adjacency matrix set of the edges where the target node is associated. I means identity matrix, and it allows the number of actual message passing steps in the searched meta structure to be flexible. O means empty matrix, and it allows the message passing steps to skip this message passing if the empty matrix receives the highest weight. They allow the number of actual message passing steps and incoming links for $\boldsymbol{H}^{(l)}$ in the searched meta structure to be flexible.

Here, we introduce a noun named search space, a set of all feasible meta structures for the heterogeneous graph, which is derived from Neural Architecture Search [23]. As demonstrated by Equation (4), the search space is influenced by three parameters: the adjacency matrix set $\mathcal{A}$, the adjacency matrix set of the edges where the target node is associated $\overline{\mathcal{A}}$, and the message passing layers L. The heterogeneous graph's diverse edge types result in an exponential increase in possible permutation and combination operations. This necessitates calculating numerous possibilities during the search, resulting in an extraordinarily vast search space. Using Amazon as an example(details in Table 1), we seek to predict customers' rates for items. The search space size is $9^3 \times (9+1)^3 \times (2)^1 \times (2+1)^3 + 9^3 \times (9+1)^3 \times (5)^1 \times (5+1)^3 \approx$

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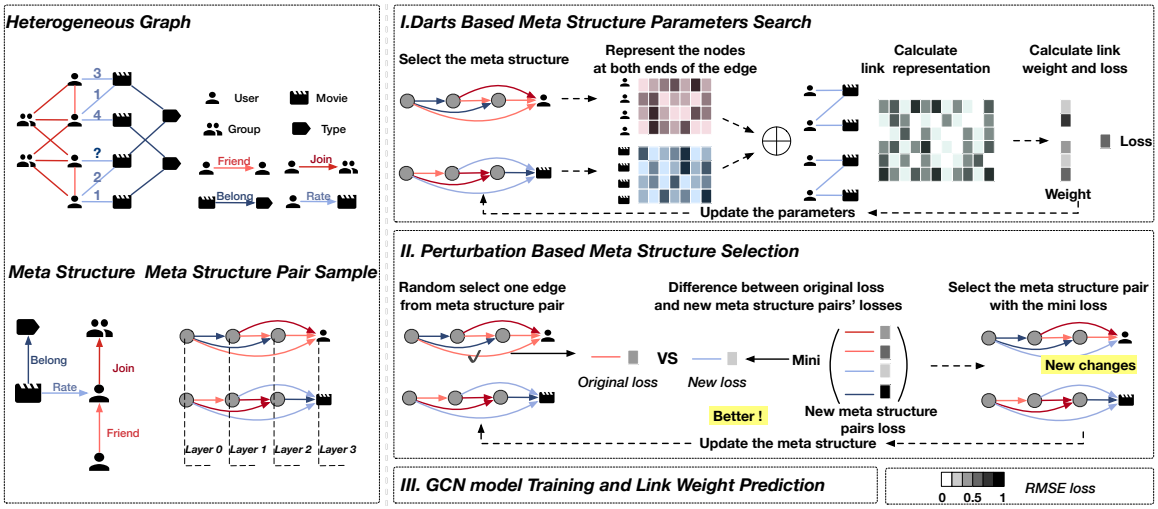


Fig. 1. The concepts and process illustration of HLWP. The left sub-graph is the heterogeneous graph, meta structure, and a meta structure pair sample. They are defined in Section 2.1. The right sub-graph is an illustration of HLWP. In part I, HLWP searches the parameters with DARTS based algorithm and gets the meta structure pair parameters (Section 3.2). In part II, HLWP selects the meta structure pair with a perturbation-based algorithm (Section 3.3). In part III, HLWP trains a GCN model and predicts link weights (Section 3.3).

1.57×10^9 with the message passing layer is four. Given the substantial magnitude of this figure, adeptly maneuvering through the extensive complexities of a large search space is imperative. In response, we use DARTS-based meta structure parameters search to mitigate this issue.

3.2. DARTS Based Meta Structure Parameters Search

In order to augment the search capacity of the meta structure, we learn from DARTS methodology, converting discrete edge types within the exploration space into continuous operations. This transformation enables training across all conceivable edge types via backpropagation, facilitating the parameter optimization of meta structures through their performance on a validation set. The utilization of gradient descent optimization ensures enhanced algorithmic efficiency, considerably reducing computational demands compared to less efficient black-box or heuristic search methods. To achieve edge-type selection continuity, we employ the subsequent equation:

$$f_{i,l}\left(\boldsymbol{H}^{(i)}, \mathcal{A}_{i,l}\right) = \sum_{A \in \mathcal{A}_{i,l}} \frac{\exp\left(\alpha_{i,l}^{A}\right)}{\sum_{A' \in \mathcal{A}_{i,l}} \exp\left(\alpha_{i,l}^{A'}\right)} \cdot \mathrm{F}\left(\boldsymbol{H}^{(i)}, A\right), \quad (5)$$

where $f_{i,l}$ denotes passing $\boldsymbol{H}^{(i)}$ message along a particular relation type given by the corresponding edge of the meta structure, $\alpha_{i,l}^{A}$ indicates the parameters of the operation corresponding to the matrix A on the edge of layer i to layer l in the meta structure, $\frac{\exp\left(\alpha_{i,l}^{A}\right)}{\sum_{A' \in \mathcal{A}_{i,l}} \exp\left(\alpha_{i,l}^{A'}\right)}$ represents the l_2 normalization of the operation corresponding to the matrix A on the edge of layer i to layer l.

Then, we need to train parameters of the meta structure pair (ms_s, ms_t) alongside parameters of GCN simultaneously. This process can be framed as a bi-level optimization problem:

$$\begin{aligned} &\min_{(\alpha_s+\alpha_t)} \ell_{\text{val}}\left(\omega^*(\alpha_s+\alpha_t), (\alpha_s+\alpha_t)\right), \\ &s.t.\ \ \omega^* = \arg\min_{\omega} \ell_{\text{tra}}(\omega, (\alpha_s+\alpha_t)), \end{aligned} \quad (6)$$

where (α_s, α_t) represents parameters of the meta structure pair (ms_s, ms_t), and ω denotes the parameters of GCN. ℓ_{val} denotes the loss on the validation set, and ℓ_{tra} denotes the loss on the training set. In other words, we minimize ℓ_{tra} by updating ω with fixed (α_s, α_t) on the training set, while minimizing ℓ_{val} by updating (α_s, α_t) with fixed ω on the validation set.

We calculate the loss of the network using Root Mean Squad Error (RMSE). Specifically, the loss function ℓ is defined as follows:

$$\ell = \sqrt{\frac{1}{|E|} \sum_{e \in E} \left(W_{pre}(e, ms_s, ms_t) - W_{label}(e)\right)^2}, \quad (7)$$

where $W_{pre}(e, ms_s, ms_t)$ indicates the predicted weight of edge e when the meta structures of the two ends nodes are ms_s and ms_t respectively.

The above process may continue for many epochs until it converges or reaches the training round. After convergence, parameters (α_s, α_t) of the meta structure pair (ms_s, ms_t) are obtained.

3.3. Perturbation Based Meta Structure Selection

After above steps, we get parameters (α_s, α_t) of the meta structure pair (ms_s, ms_t). Usually, as DARTS does, operations with maximum value parameters in (α_s, α_t) would be chosen as the final meta structure pair (ms_s, ms_t), with the implicit assumption that parameter values reflect operation strength. Previous work [24] of applying DARTS to heterogeneous graph and meta structure did precisely that. This mechanism of selecting meta structure has advantages due to its significant computational efficiency advantages. However, previous research [25] has shown that this mechanism may not reliably achieve excellent results during the selection process, resulting in architectural deterioration. This weakness is referred to as the instability of DARTS. Another study [26] has demonstrated that the maximum value parameter is not necessarily indicative for an operation's contribution to a task's overall efficacy. This paper proposes a perturbation-based selection alternative that directly measures the impact of each operation on the ultimate task.

Inspired by it, we employ a perturbation-based selection alternative when selecting the meta structure pair based on the highest performance. Specifically, we do the following steps. Firstly, we randomly select an edge (i, l) marked by two ends of nodes ID from the meta structure pair (ms_s, ms_t) and calculate the result if we skip this edge propagation. This result is used as a base for evaluation and recorded as $r_{(i,l)}$. This edge (i, l) can choose several types according to Equation (4). So, we acquire several different types of edge and calculate their predictive performance separately $r_{(i,l)_k}, k \in \mathcal{A}_{i,l}$, $\mathcal{A}_{i,l}$ is defined in Equation (4). Then, we evaluate how these results differ from the base result $r_{(i,l)}$ and

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select the meta structures pair corresponding to the most significant difference. A larger difference means a more significant influence on the result according to the proof of predecessors [26]. Then we use Equation (6) to train GCN parameters ω until it converges. After repeating the above steps for all edges, we obtain the final meta structure pair.

Finally, we use the selected meta structure pair ($m\mathcal{S}_s$, $m\mathcal{S}_t$) to train a GCN model and obtain the final link weight W_{pre}.

4. EXPERIMENT

4.1. Experimental Setup

We conducted experiments on three real-world datasets: Amazon Douban Movie , and Yelp , to evaluate our proposed method for heterogeneous graph link weight prediction. Table 1 presents the statistical information on the semantic relationships in these datasets. We discretize the link weights into several class labels based on their values. We randomly partition the links into three sets: a training set, a validation set, and a test set, with a ratio of 3:1:1. All rating classes are distributed proportionally across datasets. Experiments are conducted on a single RTX 2080 Ti GPU with 11GB memory.

We compare our HLWP against a state-of-the-art link weight prediction model LWP-WL [7], three homogeneous GNNs: node2vec [27], GCN [28] and GAT [29]; two heterogeneous GNNs based on meta paths: HGT [30] and MHGCN [31]; one heterogeneous GNN based on meta structures: DiffMG [19]. Then, we use Mean Squad Error (MSE) and Mean Absolute Error (MAE) as the measure for comparison.

Table 1. Datasets' details for Amazon, Douban and Yelp.

Dataset	Nodes	Relations	Data Split
Amazon	User(U): 6,170 Item(I): 2,753 View(V): 3,857 Brand(B): 334 Category(C): 22	U-I: 195,791 I-V: 5,694 I-C: 5,508 I-B: 2,753	Train(U-I):117,743 Val(U-I):39,158 Test(U-I):39,160
Douban	User(U): 13,367 Movie(M): 12,677 Group(G): 38 Actor(A): 2,753 Director(D): 6,311 Type(T): 2,449	U-M: 1,068,278 U-G: 570,047 U-U: 6,122 M-A: 33,572 M-D: 11,276 M-T: 27,668	Train(U-M): 640,963 Val(U-M): 213,655 Test(U-M): 213,657
Yelp	User(U): 16,239 Business(B): 14,284 Category(Ca): 512 City(Ci): 48 Compliment(Co): 12	U-B: 198,397 U-U: 158,590 U-Co: 76,875 B-Ca: 40,009 B-Ci: 14,267	Train(U-B): 119,036 Val(U-B): 39,680 Test(U-B): 39,861

4.2. Results

We use the same performance metrics with [7]. The experimental results are presented in Table 2. These findings provide the following insights. First, homogeneous GNNs regard all edges as single relations and perform badly on both datasets. This shows that neglecting semantic information in relations is risky. It is meaningful in studying heterogeneous GNNs. Second, compared with link weight prediction models, HLWP improves performances on both datasets. LWP-WL ignores semantic information and makes it hard to find useful prototypes since it considers all edges similarly. Moreover, LWP-WL picks neighborhoods based on the shortest distances, which generates a lot of IO overhead and makes the method inefficient. Third, comparing heterogeneous graph models horizontally, meta structure based heterogeneous models outperform meta path based heterogeneous models. This shows that meta structure message passing is more expressive and can aggregate node properties than meta paths. Fourth, HLWP outperforms all baselines as it can automatically utilize task-dependent semantic information via meta structure message passing. Moreover, DiffMG and HLWP perform well relative to other baselines. So we add some further experiments to evaluate the two models. We exhibit DiffMG and HLWP averages on 100 independently searched models from various search seeds to test stability. The findings demonstrate that HLWP is more stable than DiffMG.

Table 2. Experimental results on Amazon, Douban and Yelp w.r.t. RMSE and MAE. avg. is obtained on 100 independent seeds to explore the stability. The best and second-best results are shown in bold and underlined.

	Amazon		Douban		Yelp	
	MSE	MAE	MSE	MAE	MSE	MAE
node2vec	0.21830	0.06466	0.20367	0.05615	0.26291	0.08247
GCN	0.04294	0.3044	0.03417	0.03363	0.04196	0.03943
GAT	0.03154	0.02965	0.03013	0.03004	0.04025	0.03975
LWP-WL	0.04022	0.03178	0.03179	0.02339	0.03872	0.03762
HGT	0.03585	0.02826	OOM	OOM	OOM	OOM
MHGCN	0.03138	0.02084	OOM	OOM	0.03252	0.02813
DiffMG	0.00890	0.00674	0.01960	0.01569	0.02175	0.01682
HLWP(ours)	**0.00798**	**0.00634**	**0.01082**	**0.00925**	**0.01375**	**0.01104**
DiffMG(avg.)	0.01265	0.01069	0.02131	0.01796	0.02248	0.01708
HLWP(avg.)	**0.00882**	**0.00732**	**0.01119**	**0.00939**	**0.01599**	**0.01267**

4.3. Ablation Studies

We evaluate HLWP's pivotal constituents, encompassing the DARTS-based meta structure parameter search and the perturbation-based meta structure selection. This assessment offers nuanced insights into the practical components of HLWP. To assess the necessity of the DARTS based meta structure parameter search, we generate random meta structure parameters, forming HLWP$-d$. Similarly, to gauge the significance of the perturbation-based meta structure selection, we select the final meta structure pair via maximum parameter value without perturbation, resulting in HLWP$-p$. The outcomes are depicted in Figure 2. Mean scores obtained with distinct random seeds consistently demonstrate HLWP's superior performance over HLWP$-d$ and HLWP$-p$, underscoring the criticality of its core elements—DARTS based meta structure parameter search and perturbation based meta structure selection.

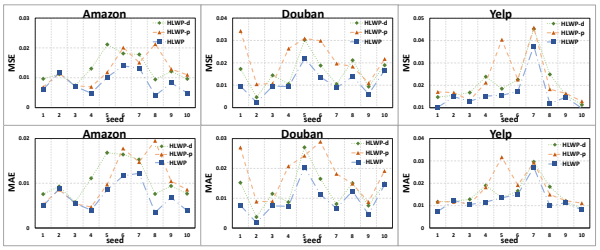


Fig. 2. Ablation study results for Amazon, Douban, and Yelp between HLWP$-d$, HLWP$-p$, and HLWP, w.r.t. RMSE and MAE.

5. CONCLUSION

This work focuses on predicting link weight on heterogeneous graphs and introduces a novel model called HLWP. HLWP employs an automatic meta structure pair generation strategy for the given heterogeneous graph, facilitating enhanced feature extraction and prediction accuracy. Experimental results on real-world datasets demonstrate the capacity of HLWP to generate innovative meta structure pairs and achieve superior link weight prediction performance. Furthermore, our research will be towards predicting link weights in dynamic heterogeneous graphs and graphs with unique attributes.

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